

EFT – Module 5

Easy-to-Read Revision Notes • Important Questions with Solutions

Exam use: These answers are written in simple university-style language. For a 5–10 mark answer, write the definition, principle, formula/derivation, working, and applications wherever applicable.

1. Explain the Cathode-Ray Oscilloscope (CRO) using electrostatic deflection.

Answer: A CRO is an electronic instrument used to display and measure electrical signals, "such as voltage, frequency, phase difference and waveform shape.

Principle: A beam of electrons is produced and accelerated inside a cathode-ray tube. "The electron beam is deflected by an electric field between pairs of deflection plates and finally strikes a fluorescent screen, producing a visible spot or waveform.

Main parts:

1. Electron gun – produces and accelerates electrons.
2. Vertical deflection plates – deflect the beam according to the input signal.
3. Horizontal deflection plates – provide horizontal/time-base deflection.
4. Fluorescent screen – converts the electron beam into visible light.

Electrostatic deflection: When a voltage is applied between the deflection plates, an electric field is produced. "The electron experiences a force $F = qE$, causing the beam to move. The displacement of the spot is approximately proportional to the applied voltage.

Applications: waveform observation, voltage measurement, frequency measurement, phase measurement and testing of electronic circuits.

2. Explain the biological effects of electromagnetic fields.

Answer: Electromagnetic fields can interact with biological tissues. The effect depends on frequency, field strength, exposure time and distance from the source.

Major effects:

- **Heating:** Radio-frequency and microwave fields can cause tissue heating because energy is absorbed by the body.
- **Nerve and muscle stimulation:** Lower-frequency fields can induce electric currents in tissues, which may stimulate nerves and muscles at sufficiently high exposure levels.
- **Eye effects:** Strong microwave/RF exposure can cause heating of sensitive tissues such as the eye.
- **Long-term exposure:** Research and safety standards evaluate possible health effects using exposure limits; effects depend strongly on frequency and exposure conditions.

Safety measures: increase distance from the source, reduce exposure time, use shielding where appropriate, maintain equipment properly and follow applicable exposure standards.

Exam tip: Draw a neat labelled diagram wherever possible. Underline the main formula and write applications at the end.

3. Explain antenna types and the mechanism of radiation.

Answer: An antenna is a device that converts guided electrical energy into electromagnetic waves during transmission and converts electromagnetic waves into electrical signals during reception.

Common antenna types:

- Dipole antenna – simple wire antenna, commonly used at VHF/UHF frequencies.
- Monopole antenna – approximately one-quarter wavelength over a ground plane.
- Loop antenna – conductor formed into a loop; useful in receiving and direction-finding applications.
- Horn antenna – flared waveguide, widely used at microwave frequencies.
- Parabolic reflector – uses a reflecting dish to obtain high directivity and gain.

Radiation mechanism: A time-varying current in the antenna produces time-varying electric and magnetic fields. These fields are coupled together and propagate outward as an electromagnetic wave. "In the far field, the electric field E and magnetic field H are mutually perpendicular and also perpendicular to the direction of propagation.

Important point: Accelerating/time-varying charges are responsible for electromagnetic radiation.

4. Explain an electrostatic generator and give its applications.

Answer: An electrostatic generator is a device that produces very high voltage using electrostatic principles. "A well-known example is the Van de Graaff generator.

Working:

1. A moving insulating belt carries electric charge upward.
2. A lower electrode supplies charge to the belt.
3. At the upper terminal, charge is transferred from the belt to a large conducting dome.
4. Charge accumulates on the outer surface of the dome, producing a very high potential.

Applications: high-voltage experiments, particle acceleration, insulation testing, electrostatic demonstrations and research involving high electric fields.

Key idea: Charge on a conductor resides on its outer surface, allowing a high potential to build up on a large isolated dome.

Exam tip: Draw a neat labelled diagram wherever possible. Underline the main formula and write applications at the end.

5. Explain magnetic deflection of charged particles.

Answer: A charged particle moving through a magnetic field experiences a magnetic force called the Lorentz magnetic force.

Formula:

$$F = q(v \times B)$$

$$\text{Magnitude: } F = qvB \sin \theta$$

where q = charge, v = particle velocity and B = magnetic flux density.

If the particle enters the magnetic field perpendicular to the field, $\theta = 90^\circ$, so:

$$F = qvB$$

The force is perpendicular to the velocity, so it changes the direction of motion but not the speed. The particle therefore follows a circular path.

For circular motion:

$$qvB = mv^2/r$$

Therefore, $r = mv/(qB)$.

Applications: mass spectrometers, particle accelerators, magnetic spectrographs and charged-particle beam control.

6. Explain the Hall effect and derive Hall voltage.

Answer: The Hall effect is the development of a transverse voltage across a current-carrying conductor or semiconductor when it is placed in a magnetic field perpendicular to the direction of current.

Suppose current flows along x-direction and magnetic field B is along z-direction. Charge carriers experience a magnetic force:

$$F_B = qv_d B$$

Charges accumulate on one side and produce a Hall electric field E_H . At equilibrium:

$$qE_H = qv_d B$$

Therefore, $E_H = v_d B$.

If the Hall voltage is measured across width w:

$$V_H = E_H w = v_d B w$$

Using $I = nqAv_d$ and $A = wt$:

$$v_d = I/(nqwt)$$

Hence:

$$V_H = BI/(nqt)$$

where n is carrier concentration and t is specimen thickness.

Applications: magnetic-field measurement, current sensing, position/speed sensing and identifying the type of charge carrier.

Exam tip: Draw a neat labelled diagram wherever possible. Underline the main formula and write applications at the end.

7. Explain the working of an electromagnetic pump.

Answer: An electromagnetic pump moves a conducting liquid, such as liquid metal, without using mechanical moving parts.

Principle: It works on the magnetic force experienced by a current-carrying conductor in a magnetic field.

A current is passed through the conducting liquid while a magnetic field is applied perpendicular to the current. The liquid experiences a force:

$\mathbf{F} = \mathbf{J} \times \mathbf{B}$ per unit volume, where J is current density and B is magnetic flux density.

This force pushes the conducting liquid through the pipe or channel.

Advantages: no mechanical impeller, low mechanical wear, suitable for high-temperature conducting

fluids and relatively simple construction.

Applications: liquid-metal circulation, metallurgy, cooling systems and specialized high-temperature pumping systems.

8. Explain electromagnetic interference (EMI) and methods of reducing it.

Answer: Electromagnetic interference (EMI) is unwanted electromagnetic energy that disturbs the normal operation of an electronic or electrical system.

Sources of EMI: motors, switching circuits, power supplies, radio transmitters, digital circuits, ignition systems and other rapidly changing currents/voltages.

Coupling mechanisms:

- Conducted interference – travels through wires or power lines.
- Radiated interference – travels through space as electromagnetic radiation.
- Capacitive coupling – electric-field coupling between nearby conductors.
- Inductive coupling – magnetic-field coupling due to changing current.

EMI reduction techniques:

1. Shielding – use conductive enclosures or shields.
2. Grounding – provide proper low-impedance grounding paths.
3. Filtering – use capacitors, inductors and EMI filters to suppress unwanted frequencies.
4. Twisted-pair cables – reduce magnetic pickup and radiation.
5. Proper PCB layout – minimize loop areas and keep noisy/high-speed sections separated from sensitive circuits.
6. Ferrite beads/cores – suppress high-frequency noise on cables.

Goal: electromagnetic compatibility (EMC) means equipment should operate correctly without causing unacceptable interference to other equipment.

Exam tip: Draw a neat labelled diagram wherever possible. Underline the main formula and write applications at the end.

Quick Formula Sheet

Topic	Formula / Key relation
CRO / electrostatic deflection	$F = qE$
Magnetic force	$F = q(\mathbf{v} \times \mathbf{B})$, $ F = qvB \sin\theta$
Circular motion in B-field	$r = mv/(qB)$
Hall effect	$V_H = BI/(nqt)$
Electromagnetic pump	Force density $\approx \mathbf{J} \times \mathbf{B}$
Electric field	$E = F/q$
Electric flux density	$D = \epsilon E$
Poynting vector	$\mathbf{S} = \mathbf{E} \times \mathbf{H}$
Wave relation	$v = 1/\sqrt{\mu\epsilon}$
Free-space impedance	$\eta_0 = \sqrt{(\mu_0/\epsilon_0)} \approx 377 \Omega$

Last-Minute Answer Structure

For 5 marks: Definition → Principle → 2–4 key points → Formula/diagram → Applications.

For 10 marks: Definition → Principle → labelled diagram → step-by-step working/derivation → formula → applications → conclusion.

For numericals: Given → Required → Formula → Substitution with units → Final answer with unit.